

RESEARCH ARTICLE

Giant-Exchange-Driven Vectorial Control of a Minimal Topological Magnet in $\text{Eu}_3\text{In}_2\text{As}_4$

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ABSTRACT

The interplay between magnetism and band topology provides a route to controlling quantum states of matter, yet its realization in materials is often constrained by weak exchange coupling and complex electronic structures. Here, a giant exchange coupling is identified in the newly predicted topological magnet $\text{Eu}_3\text{In}_2\text{As}_4$, giving rise to magnetization-dependent band shifts of up to 300 meV. Together with its intrinsically soft magnetic response, this strong coupling enables systematic tuning of topological phases by both the magnitude and orientation of applied magnetic fields. The magneto-topological phase diagram is mapped out in which an antiferromagnetic topological insulator ground state evolves, under modest fields, into a proposed intermediate 2/3-ferrimagnetic phase, and further into fully polarized ferromagnetic states predicted to host either Weyl or nodal-ring semimetals. Notably, the Weyl phase corresponds to a minimal model hosting a single pair of Weyl nodes. Quantum oscillations, anomalous Hall transport and magneto-infrared spectroscopy consistently reveal exchange-driven band reconstruction across these transitions. Rotation of the magnetization theoretically provides an efficient means to tune the momentum-space positions and separations of the Weyl nodes. These results establish $\text{Eu}_3\text{In}_2\text{As}_4$ as a model system for exploring how strong exchange coupling can be used to control topological band structures with minimal complexity.

1 | Introduction

The integration of intrinsic magnetism with quantum materials offers a powerful avenue for realizing and dynamically control-

ling diverse exotic topological phases [1–3]. Central to this synergy is the exchange coupling between localized magnetic moments and itinerant electrons, which breaks time-reversal symmetry (TRS) and profoundly reshapes electronic band structures. In

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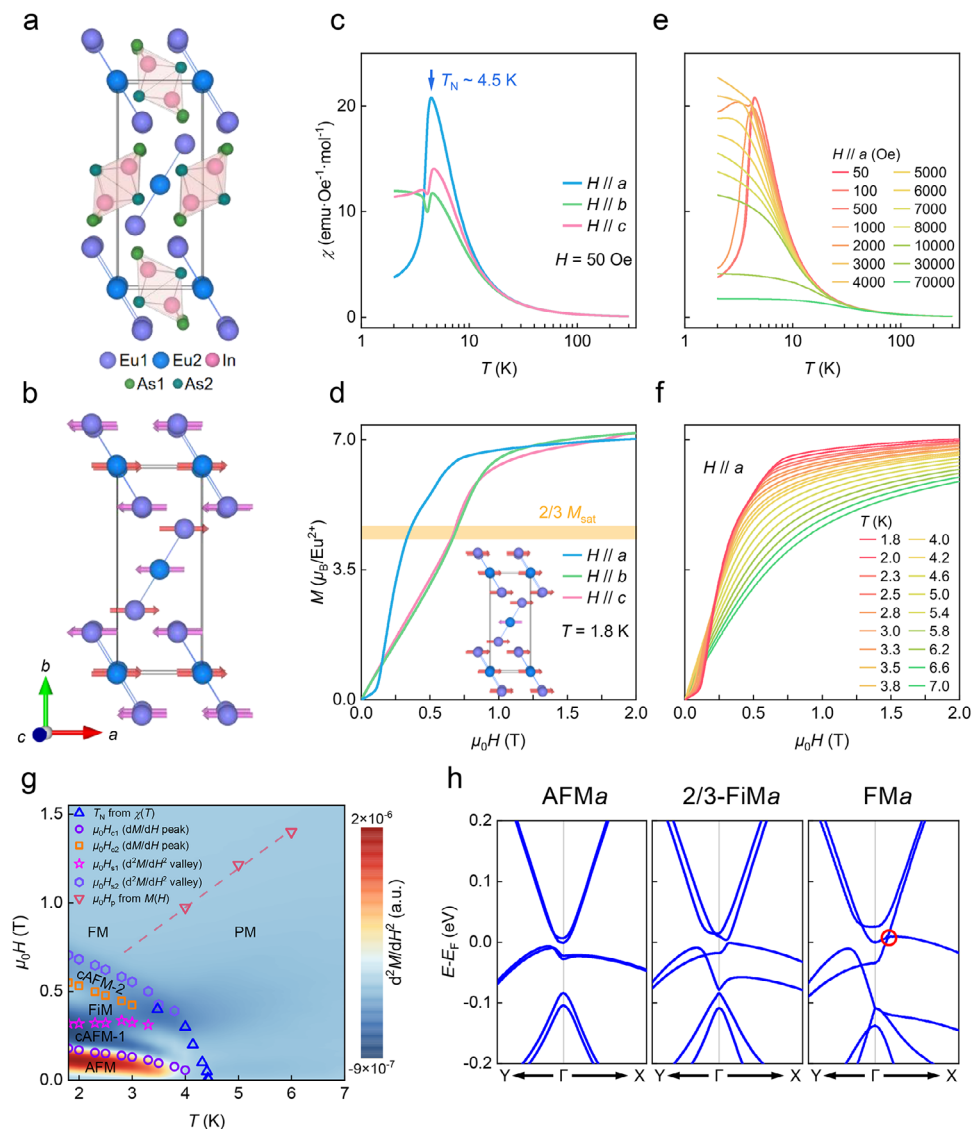


FIGURE 1 | Crystal structure and magnetism of $\text{Eu}_3\text{In}_2\text{As}_4$. (a) Schematic crystal structure of $\text{Eu}_3\text{In}_2\text{As}_4$. The inequivalent Eu sublattices are represented by purple and blue atoms. (b) Magnetic structure of the C-type AFMa ground state, forming FM coupling along c -axis and AFM coupling within the ab -plane. For clarity, only Eu atoms are shown here. Colored arrows denote oppositely oriented spins, with the two Eu sublattices each forming AFM structures. (c) Temperature dependence of magnetic susceptibility measured under field-cooled (FC) conditions with an applied field of 50 Oe along the three principal crystallographic axes. (d) Field-dependent magnetization at 1.8 K with the field applied along three crystallographic axes. The yellow shaded region denotes the $2/3 M_{\text{sat}}$ regime, with its upper and lower boundaries defined by two-thirds of the magnetization measured at 7 T and at the ferromagnetic saturation field, respectively. (e) Temperature-dependent susceptibility measured under increasing magnetic field along a -axis. (f) Field-dependent magnetization at various temperatures with the field applied along a -axis. (g) Magnetic phase diagram of $\text{Eu}_3\text{In}_2\text{As}_4$ for $H \parallel a$. The color plot background exhibits the evolution of d^2M/dH^2 with field and temperature. Features in the second derivative evidence the $2/3$ -FiMa state. The red dashed line is a guide to the eye denoting the FM-PM phase boundary. (h) Band structures of the AFMa, $2/3$ -FiMa, and FMa states (from left to right, respectively) along $\text{Y}-\Gamma$ path, illustrating a continuous bandgap closing process.

magnetic topological insulators, such coupling opens gaps in surface Dirac cones, enabling axion electrodynamics and the quantum anomalous Hall effect [1, 2, 4–6]. In Weyl semimetals, TRS breaking generates pairs of Weyl nodes with opposite chirality, whose separation in momentum space governs large intrinsic anomalous Hall effect (AHE) [7–10] and anomalous Nernst effect (ANE) [11–14]. The strength and tunability of the exchange coupling thus dictate not only the stability of these phases but also the degree to which they can be manipulated by external fields, a prerequisite for both fundamental exploration and device applications.

Despite significant progress, existing magnetic topological materials face persistent challenges that limit their utility. Many candidates exhibit complex band structures riddled with multiple bands near the Fermi level, obscuring the clean realization of minimal topological models [10, 13, 15–22]. Moreover, the exchange coupling between local moment and itinerant carriers in most systems is weak [15, 23, 24], typically yielding band shifts on the order of a few to tens of meV, rendering topological transitions either inaccessible or requiring extreme conditions. Compounding these issues, low carrier mobility often masks intrinsic topological transport signatures [22, 25–29], while

strong magnetic anisotropy restricts field-tunability to narrow parameter windows. These limitations underscore the urgent need for new platforms where strong exchange couplings coexist with simple, well-isolated topological bands and soft magnetic response.

Here, we introduce $\text{Eu}_3\text{In}_2\text{As}_4$ as a model magnetic topological material that overcomes these limitations. We uncover a giant exchange coupling—manifested as magnetization-dependent band shifts up to 300 meV, far exceeding conventional Zeeman scales, which drives pronounced reconstructions of the electronic structure across multiple magnetic phases. $\text{Eu}_3\text{In}_2\text{As}_4$ hosts an antiferromagnetic (AFM) topological insulator ground state below the Néel temperature ($T_N = 4.5$ K), which evolves under modest magnetic fields into a 2/3-ferrimagnetic intermediate phase, and ultimately into fully polarized ferromagnetic (FM) states predicted to realize ideal Weyl or nodal-ring semimetals. Remarkably, these FM states feature the minimal Weyl model: a single pair of Weyl nodes, free from extraneous band complications. Owing to intrinsic magnetic softness, full polarization is achieved with fields as low as ~ 1 T along any crystallographic axis, and rotation of the magnetization direction is proposed to efficiently tune the momentum-space positions and separations of the Weyl nodes. Through quantum oscillations, anomalous Hall transport, and magneto-infrared spectroscopy, we provide consistent evidence of exchange-driven band reconstruction and topological phase transitions. Our work establishes $\text{Eu}_3\text{In}_2\text{As}_4$ as a uniquely versatile platform where giant exchange coupling enables unprecedented magnetic control over clean, tunable, and minimal topological band structures.

2 | Results

2.1 | Diverse Magnetic Phases With Distinct Topological Band Structures

$\text{Eu}_3\text{In}_2\text{As}_4$ crystallizes in the orthorhombic space group $Pn\bar{m}$ (No. 58) [30], adopting a Zintl-phase architecture composed of quasi-one-dimensional $[\text{In}_2\text{As}_4]$ chains extending along the c -axis, intercalated by two inequivalent Eu^{2+} sites in the ab -plane (Figure 1a). Single crystals were structurally and compositionally characterized via x-ray diffraction (XRD), energy-dispersive x-ray spectroscopy (EDX) and Raman spectrum (see Notes S1 and S2). Energy calculations identify the AFM state with the Néel vector along a -axis (AFM a) as the magnetic ground state (energy differences listed in Table S3). In this phase, Eu^{2+} moments form a C-type AFM order, where the two inequivalent Eu^{2+} sites constitute two AFM sublattices respectively (Figure 1b).

Magnetic susceptibility $\chi(T)$ measurements (Figure 1c) confirmed the AFM ground state, revealing peaks at $T_N \sim 4.5$ K for all field orientations. Below T_N , $\chi(T)$ for $H // a$ sharply drops toward zero, whereas for fields perpendicular to a -axis, lower transition peaks and low-temperature upturns are observed. This anisotropy indicates the a -axis as the easy magnetization direction. Curie–Weiss fitting (Figure S5) in the paramagnetic (PM) regime yields positive Weiss temperatures for all axes, suggesting short-range FM correlations between Eu^{2+} ions above T_N , despite the AFM ground state. Such contrast, previously reported in low-carrier-density Eu^{2+} -based antiferromagnets, is attributed

to magnetic polaron formation [31–37]. These polarons arise from strong exchange coupling between localized $4f$ moments and itinerant carriers, leading to FM clusters embedded in a PM or AFM matrix [38]. Beyond the positive Weiss temperatures, the low-temperature upturns in $\chi(T)$ for $H // b$ and c which signal weak ferromagnetism and the deviation from the Curie–Weiss law support this scenario of magnetic polarons (Figure S5).

Figure 1d exhibits the magnetization $M(H)$ curves at 1.8 K. For $H // a$, a low-field plateau below 0.1 T reflects the AFM easy-axis configuration, followed by a sharp magnetization jump resembling a spin-flop transition and an additional anomaly near 0.5 T. In contrast, for $H // b$ or c , M initially increases linearly at low fields and exhibits slope changes around 0.69 T before gradually approaching saturation. These features are more clearly resolved in dM/dH and d^2M/dH^2 curves. Unlike conventional easy-axis antiferromagnets with weak anisotropy, which typically exhibit a single spin-flop transition, $\text{Eu}_3\text{In}_2\text{As}_4$ displays two successive sets of anomalies in both dM/dH and d^2M/dH^2 curves for $H // a$ (Figure S6), indicating two consecutive field-induced reorientation/polarization processes rather than a single-step transition.

$\text{Eu}_3\text{In}_2\text{As}_4$ contains two crystallographically inequivalent Eu sublattices (Eu1 and Eu2) with multiplicities of four and two per magnetic unit cell, contributing 2/3 and 1/3 of the total saturated moment, respectively. We therefore propose a sequential polarization scenario: starting from the AFM a ground state, the Eu1 sublattice first undergoes a spin-flop transition and evolves into a canted AFM (cAFM-1) state. As the field increases, the Eu1 moments continuously cant toward a -axis and reach full polarization first, yielding an intermediate ferrimagnetic-like state with approximately 2/3 of the saturated magnetization (denoted as 2/3-FiMa), while Eu2 sublattices remain AFM-coupled (Figure 1d inset). The Eu2 sublattice subsequently undergoes its own spin-flop and polarization process (cAFM-2), ultimately driving the system into the fully polarized FM a state. This sequential evolution is schematically summarized in Figure S7. Energy calculations support the energetic feasibility of the 2/3-FiMa configuration as an intermediate state between AFM a and FM a states (Table S3). Analogous consecutive magnetization on different Eu-sites was reported in Zintl antiferromagnet $\text{Eu}_5\text{In}_2\text{As}_6$ [31].

For $H // b$ and c , Eu1 spins continuously rotate from the easy a -axis toward the field direction until alignment. The peak features in dM/dH again occur when the magnetization reaches approximately 2/3 of the saturated value, consistent with the sequential polarization scenario proposed above. Beyond this field, the remaining Eu2 sublattice progressively aligns with the external field, inducing an increase of M with a more rapid slope until complete spin polarization is achieved. At high fields, M saturates first along the easy a -axis, and a small field range (~ 1 T) is sufficient to achieve full spin polarization for all axes, revealing soft magnetism and weakly-coupled AFM ground state (also evidenced by low T_N and tiny AFM-FM energy difference). The saturated magnetization (M_{sat}) reaches $\sim 7 \mu_B/\text{Eu}^{2+}$, confirming the Eu^{2+} -dominated magnetism with half-filled $4f$ electrons ($S = 7/2$). $M(H)$ curves measured up to ± 7 T with rising temperatures are summarized in Figure S8.

Detailed static magnetization characterizations delineate magnetic phase diagram for $H\parallel a$ in Figure 1g, with the phase boundaries determined from $\chi(T)$, $M(H)$, dM/dH , and d^2M/dH^2 curves (see Note S3 for details). Higher fields suppress and shift the $\chi(T)$ peaks to lower temperatures until vanishing (Figure 1e), defining the boundary between the long-range ordered and PM states. At low temperatures, based on the sequential polarization process described above, the ordered region is divided into the AFMa, cAFM-1, 2/3-FiMa, cAFM-2, and FMa phases (Figure 1f). The characteristic fields μ_0H_c and μ_0H_s are extracted from the peaks in dM/dH and the minima in d^2M/dH^2 , respectively [39–42], and are used to divide the phase boundary. Here, μ_0H_c denotes the characteristic field of the spin-flop process, whereas μ_0H_s marks the onset of the fully polarized ferromagnetic state. Notably, the intermediate 2/3-FiMa phase emerges only at lower temperatures and occupies a finite field window near the μ_0H_{s1} line. Its microscopic spin configuration is inferred from magnetization and derivative analyses, while a definitive determination would require neutron diffraction or related magnetic-structure probes.

At elevated temperatures, thermal fluctuations suppress the inflection features and broaden the saturation process in $M(H)$, rendering the minima in d^2M/dH^2 less sharply defined, toward S-shapes in PM regime above T_N (Figure 1f). In this regime, the fields μ_0H_p for full polarization are derived from the intersections of linear fits to the low-field and high-field regions of $M(H)$, following the procedures used in Refs. [43, 44]. This provides a consistent criterion for tracing the high-temperature boundary of the field-polarized regime. Additional magnetization characterizations are detailed in Note S3.

Density functional theory (DFT) calculations incorporating spin-orbit coupling (SOC) reveal the magnetic-order-driven topological phase evolution in $\text{Eu}_3\text{In}_2\text{As}_4$ (Figure 1h). In the AFMa state, SOC reduces the indirect gap near Γ and induces band inversion, giving rise to a topologically non-trivial bandgap. Based on the reported topological surface state calculations, AFMa $\text{Eu}_3\text{In}_2\text{As}_4$ is identified as an axion insulator candidate with hybrid-order band topology [45, 46]: magnetism gaps the Dirac surface states of bc - and ac -planes, whereas the preserved C_{2z} symmetry protects an unpinning Dirac cone on the ab -plane, introducing emergent one-dimensional (1D) hinge states among the gapped surfaces. For 2/3-FiMa state, the bulk bandgap is further reduced; the remaining small gap suggests proximity to the phase boundary between a topological insulator and a semimetal, where the analysis of surface states is theoretically consistent with axion-insulator characteristics (details in Figure S12).

Most strikingly, full spin polarization transforms $\text{Eu}_3\text{In}_2\text{As}_4$ into a minimal Weyl semimetal. For FM alignment along a (or b), band inversion generates a single pair of Weyl nodes along the Γ -X (Γ -Y) path, with no additional trivial bands crossing the Fermi level, fulfilling the criteria for an ideal Weyl semimetal. When magnetization is forced along the c -axis, the system instead hosts an ideal nodal-ring within the k_x - k_y plane, stabilized by mirror symmetry (band structures for FMb/ c are detailed in Figure S11). This direct linkage between spin orientation and topological band geometry establishes $\text{Eu}_3\text{In}_2\text{As}_4$ as a uniquely programmable platform for exploring clean, exchange-driven Weyl physics.

2.2 | Magneto-Infrared Spectroscopy of $\text{Eu}_3\text{In}_2\text{As}_4$

To experimentally track the evolution of the electronic structure across these distinct magnetic orders, we employed magneto-infrared spectroscopy, which enables direct monitoring of field-dependent optical transitions over a broad energy window. Magneto-reflectivity R_B measurements were performed on the (100) surface of $\text{Eu}_3\text{In}_2\text{As}_4$ at around 6 K, with the field applied along a -axis (c -axis) for Faraday (Voigt) geometry. The R_B spectra were normalized by the zero-field reflectivity R_0 , yielding relative magneto-reflectivity R_B/R_0 , presented as stacked plots (vertically constant offset) in Figure 2a (Faraday geometry) and Figure 2b (Voigt geometry). Spectral peaks were assigned to inter-band optical transitions, and three transitions, labeled T_α , T_β , and T_γ , are observed in both geometries. Moreover, the first derivatives $d(R_B/R_0)/d\omega$ were also plotted in Figure 2c–d to better resolve the peak positions and their field evolution.

As the magnetic field increases, the transition energies display a distinctly nonlinear evolution (details in Figure S13), characterized by a rapid initial change followed by gradual saturation. This behavior closely tracks the field-dependent magnetization measured at the same temperature. Although experiments were conducted above T_N due to setup limitations, the field-induced fully polarized state is still accessible at high fields, likely facilitated by strong spin fluctuations or short-range FM correlations, consistent with the positive Weiss temperatures. Importantly, the exceptionally large energy shifts—exceeding 300 meV (for T_β) within 3 T—are far beyond the scale expected for Zeeman splitting [47–52], thereby unequivocally ruling out the Zeeman effect as the underlying mechanism. Instead, these results demonstrate a highly efficient and giant band modulation related to the magnetization.

Similar M -dependent optical transitions have been reported in several Eu^{2+} -based semiconductors, including EuTe and EuCd_2X_2 ($X = \text{P, As, and Sb}$) [53–56], where they originate from exchange couplings between localized Eu^{2+} moments and itinerant electrons. We attribute the giant energy shifts observed in $\text{Eu}_3\text{In}_2\text{As}_4$ to the same mechanism.

Microscopically, Eu^{2+} possesses a half-filled $4f^7$ configuration with $S = 7/2$ and $L = 0$, giving rise to a large spin-only magnetic moment of $\sim 7 \mu_B$. Upon magnetic polarization, these localized moments generate a strong internal exchange field acting on the itinerant carriers. Within the spin-only Heisenberg picture, the exchange coupling mechanism of band splitting can be expressed as [55, 57–59]:

$$H_{\text{ex}} = -2J_{\text{ex}} \sum_i \mathbf{s}_i \cdot \mathbf{S}_i \quad (1)$$

where $\mathbf{s}_i$ and $\mathbf{S}_i$ denote the spins of itinerant electrons and neighboring Eu^{2+} moments, respectively, and exchange constant J_{ex} characterizes the coupling strength. Under the mean-field approximation, the average local Eu-spin polarization scales with the magnetization $M(T, B)$, according to $\langle \mathbf{S}_i \rangle = S \cdot M(T, B)/M_{\text{sat}}$ [55], yielding an exchange-induced band splitting energy:

$$E_{\text{ex}} = -2J_{\text{ex}} \cdot \frac{1}{2} \cdot S \cdot M(T, B) / M_{\text{sat}} = -J_{\text{ex}} S M(T, B) / M_{\text{sat}} \quad (2)$$

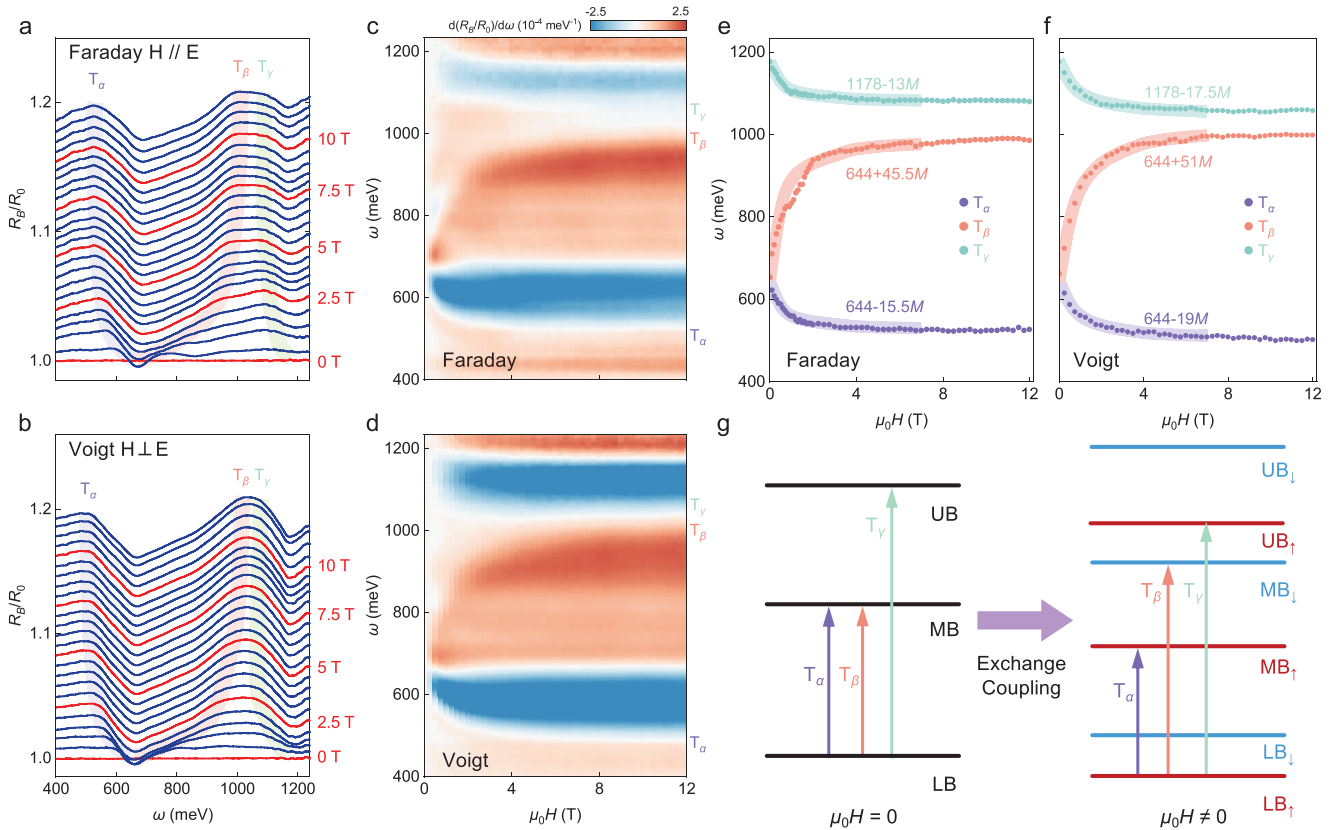


FIGURE 2 | Magneto-infrared spectra of $\text{Eu}_3\text{In}_2\text{As}_4$. (a, b) Stacked plots of relative magneto-reflectivity R_B/R_0 spectra (0.5 T step) measured in Faraday ($H // a$) and Voigt ($H // c$) geometries, respectively, with the infrared beam focused on the (100) plane. Three distinct sets of peak features, denoted T_α , T_β , and T_γ , appear and evolve with the increasing field, and are respectively traced by violet, red and green curves. (c, d) False-color plots of the first derivative of relative magneto-reflectivity $d(R_B/R_0)/d\omega$, in (c) Faraday and (d) Voigt configurations, respectively. (e, f) Correlation between magnetization and optical transition energies with Faraday and Voigt configurations in (e) and (f), respectively. Colored dots denote the apparent energies of T_α , T_β , and T_γ . Shaded curves indicate the linear scaling of magnetization curves around 6 K, with scaling parameters labeled near each curve. (g) Possible origin of optical transitions. Solid arrows represent the optical transitions predicted by the model. Black lines indicate the spin-degenerate bands, while red and blue distinguish spin-polarized bands under exchange-induced splitting.

The corresponding band shift energy of a single spin branch is $E_{\text{ex}}/2$. Because the observed infrared transition involves two different electronic bands, the measured transition-energy shift reflects the combined exchange responses of both the initial and final states. We therefore introduce an effective exchange constant J_{eff} , representing the net exchange contributions of the two participating bands. The field dependence of the transition energy can then be written as [53, 54]:

$$\Delta E = E(B) - E_0 = -\frac{1}{2}J_{\text{eff}}SM(T, B)/M_{\text{sat}} \quad (3)$$

where $E(B)$ and E_0 are the transition energies at finite and zero field, respectively. This relation predicts a linear scaling of transition energies with magnetization rather than with magnetic field itself. The extracted transition energies are plotted in Figure 2e–f for both geometries, where the field-dependent energy shifts of T_α , T_β , and T_γ are well captured by linear scaling with magnetization (solid shaded curves). Notably, the near-identical extrapolation to zero field of those three transitions in both geometries reveals a common band origin. Slight differences in scaling parameters between Faraday (Figure 2e) and Voigt (Figure 2f) geometries indicate minor exchange anisotropy,

consistent with the soft and nearly isotropic magnetism of FM- $\text{Eu}_3\text{In}_2\text{As}_4$.

Additionally, a clear decrease was observed in relative magneto-reflectivity within the 600–800 cm^{-1} range, likely associated with a redshift of the plasma frequency (Figure S14). Figure 2g illustrates a simplified model of the optical transitions. At zero field, spin-degenerate bands (in black) give rise to only two transitions: T_α and T_β originate from the lower band (LB) to the middle band (MB), while T_γ likely corresponds to a transition from the LB to the upper band (UB). With increasing field, exchange-induced spin splitting shifts the spin-up bands (red) to lower energy and the spin-down bands (blue) to higher energy. Owing to the difference in the exchange splitting scales among the relevant bands, the transition energies respond differently to magnetization: T_α and T_γ redshift with increasing magnetization, whereas T_β blueshifts. Within this framework, the exchange splitting parameters can be quantitatively extracted, yielding a giant band splitting approaching ~ 500 meV for MB (at 12 T) (see Tables S4 and S5).

Other transitions anticipated within this level scheme are not experimentally resolved (see Figure S15), as a natural

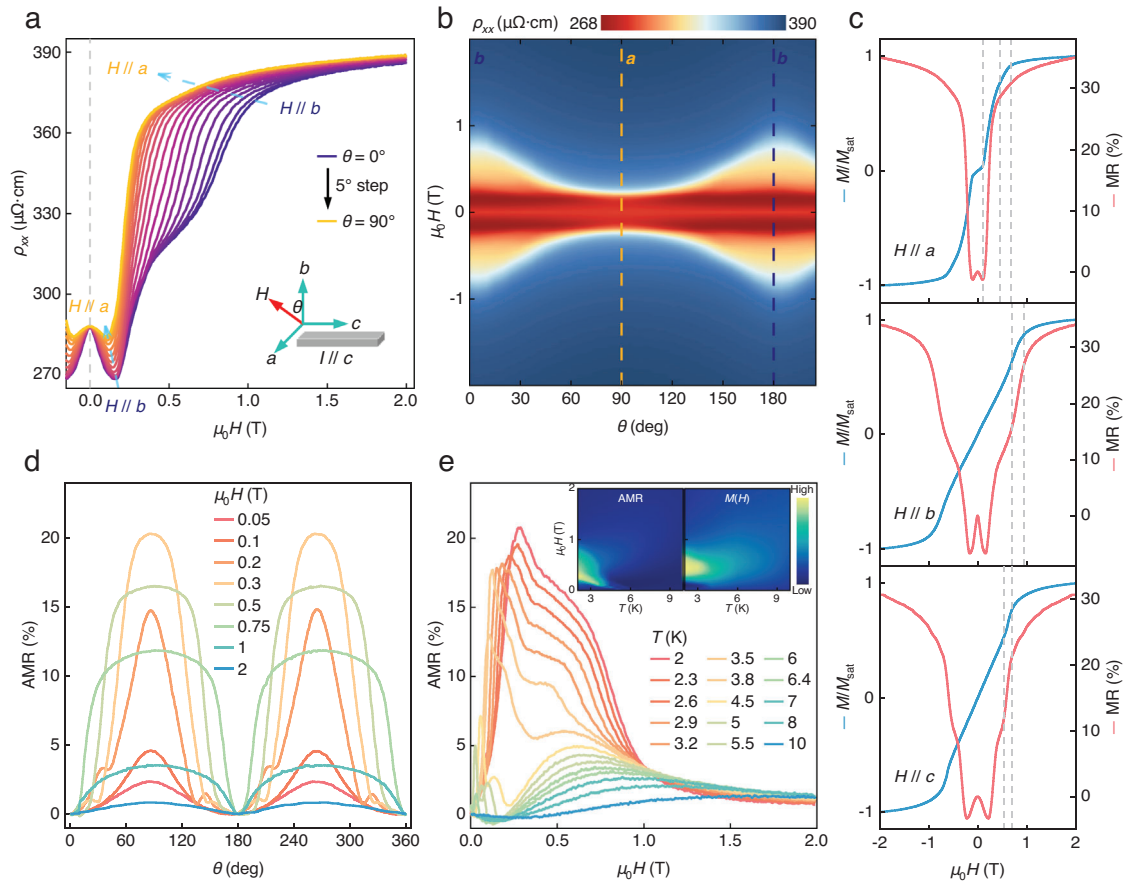


FIGURE 3 | Anisotropic magneto-transport in the low-field regime. (a) Field dependence of the magneto-resistivity, $\rho_{xx}(H)$, at 2 K. Current was applied along the *c*-axis while the field direction was rotated from the *b*- to *a*-axis by angle θ , as illustrated in the inset. (b) Color plot of the angular-dependent $\rho_{xx}(\theta, H)$, revealing *ab*-plane anisotropy. (c) Comparison between the magnetoresistance ($MR = \frac{[\rho_{xx}(H) - \rho_{xx}(0\text{ T})]}{\rho_{xx}(0\text{ T})} \times 100\%$) and magnetization curves (normalized by the saturation magnetization M_{sat}) at 2 K for field along the three primary axes. Anomalies in the magnetization can be reflected by features in the MR curves. (d) Angular-dependent MR (AMR) at selected fields (2 K), with the field rotated from 0° to 360° within the *ab*-plane. The AMR shows a two-fold symmetry with maxima at 90° (*H* // *a*). (e) Field dependence of AMR at 90° at increasing temperatures. The AMR shows a non-monotonic field dependence, peaking around 0.3 T and decreasing toward zero at higher fields. This non-monotonic behavior gradually diminishes as temperature increases above T_N . The insets show the color plot of the AMR(*H*, *T*) at 90° and magnetization difference between *a*- and *b*-axis ($M_a(H, T) - M_b(H, T)$), revealing a similar trend.

consequence of phase-space constraints and instrumental boundaries. In particular, transitions originating from $\text{LB}_{\downarrow}$ are suppressed by Pauli blocking, while higher-energy channels such as $\text{LB}_{\uparrow} \rightarrow \text{UB}_{\downarrow}$ lie beyond the detection window of the InSb detector. Although the schematic model neglects SOC and anti-ferromagnetic exchange, these effects are expected to renormalize the detailed level structure without altering the overarching physical picture. Crucially, the extracted exchange energy scale is sufficiently large to drive gap closure and band inversion in the calculated fully spin-polarized phase, thereby providing a direct experimental basis for exchange-driven low-energy band reconstruction and topological phase transitions.

2.3 | Anisotropic Magneto-Transport of Eu₃In₂As₄

To access the low-energy electronic states that govern charge transport across different magnetic orders, we therefore carried out anisotropic magneto-transport measurements on Eu₃In₂As₄. Eu₃In₂As₄ exhibits metallic behavior upon cooling, with a resis-

tivity anomaly near T_N (see Figure S16). Figure 3a displays the field-dependent resistivity $\rho_{xx}(H)$ measured up to 2 T, with current applied along the *c*-axis and field rotated within *ab*-plane by angle θ (inset, Figure 3a). The in-plane magnetoresistance (MR) anisotropy is further visualized in the angular-resolved $\rho_{xx}(\theta, H)$ mapping (Figure 3b).

For all field orientations, $\rho_{xx}(H)$ initially decreases, reaches a minimum at a critical field, and subsequently increases as field-forced FM states develop. Band calculations reveal a small bandgap in AFM_a ground state, which continuously closes upon transitioning to gapless FM semimetals, likely contributing to the initial negative magnetoresistance (NMR). Alternatively, the initial NMR may originate from a field-driven percolation transition of magnetic polarons, leading to carrier delocalization and thus reduced resistivity [35]. With further increasing the field, the MR changes sign and becomes positive before complete spin-polarization. As shown in Figure 3c, where we compare $\rho_{xx}(H)$ and $M(H)$ along three crystalline axes, the resistivity features closely track the magnetization steps (indicated by gray dashed

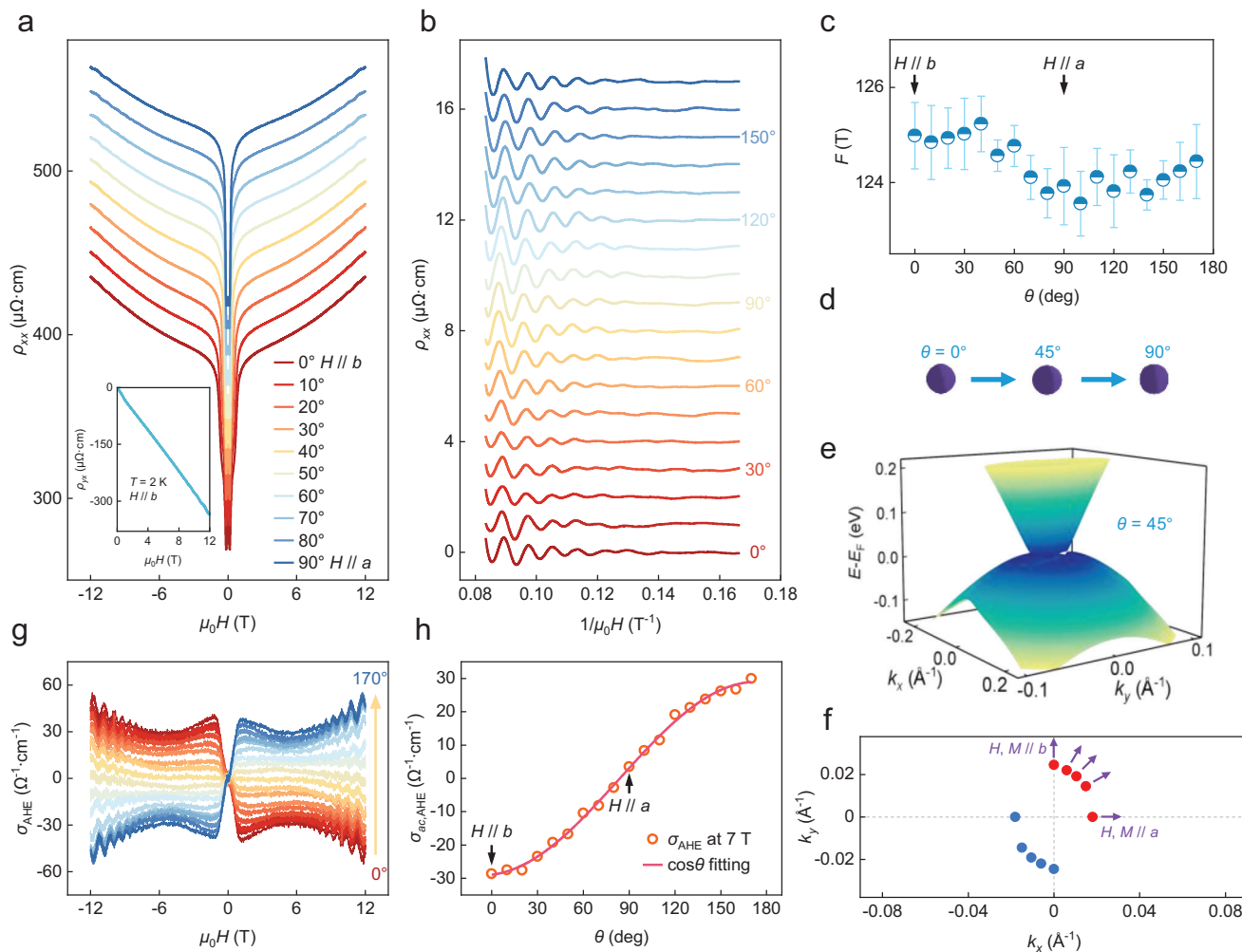


FIGURE 4 | Shubnikov-de Haas (SdH) oscillations and anomalous Hall effect (AHE) in FM-Eu₃In₂As₄. (a) Magneto-resistivity $\rho_{xx}(H)$ at 2 K within ± 12 T, with H rotated within ab -plane. Current was applied along c -axis. The $\rho_{xx}(H)$ curves are offset for clarity. Inset: Hall resistivity $\rho_{yx}(H)$ at 0°, exhibiting a nearly linear field dependence with negative slope, indicating electron conduction. Linear fitting of $\rho_{yx}(H)$ in FM state yields an electron density of $2.1 \times 10^{19} \text{ cm}^{-3}$. (b) SdH oscillations $\Delta\rho_{xx}$ (obtained by polynomial fitting) as a function of $1/\mu_0 H$ with varying field directions. (c) SdH oscillation frequencies (with error bars) versus field direction θ , showing weak angular dependence. The frequency was extracted as linear-fit slope of the Landau index n versus $1/\mu_0 H$ in Landau fan diagrams. (d) Calculated 3D Fermi surfaces of Eu₃In₂As₄ for three representative magnetization directions ($\theta = 0^\circ$, 45° , and 90°). For each configuration, the viewing direction is aligned with the applied magnetic field, so that the projected cross section corresponds to the extremal orbit perpendicular to H . The Fermi surface is plotted at $E - E_F = 0.155 \text{ eV}$, chosen to match the experimentally extracted SdH frequency. (e) 3D band structure of FM Eu₃In₂As₄ with $\theta = 45^\circ$, demonstrating a single pair of Weyl points located in the k_x - k_y plane. For clarity, only the two bands forming the Weyl crossing are shown. (f) Position evolution of single Weyl pairs of FM Eu₃In₂As₄, with the field (magnetization) is rotated in the ab -plane ($\theta = 0^\circ$, 30° , 45° , 60° , and 90°), denoted as the purple arrows. Blue and red colors represent opposite chirality of Weyl points. (g) Field-dependent anomalous Hall conductivity σ_{AHE} at 2 K with rotating H , where $\sigma_{\text{AHE}} = \rho_{\text{AHE}}/(\rho_{xx}^2 + \rho_{yy}^2)$ and $\rho_{\text{AHE}} = \rho_{yx} - R_H \mu_0 H$ (R_H is the linear Hall coefficient in the FM state). (h) Angular-dependent σ_{AHE} values (the $\sigma_{ac, \text{AHE}}$ component) at 7 T, exhibiting a $\cos\theta$ trend.

lines), demonstrating that the transport response is strongly affected by the field-induced evolution of the magnetic state.

Upon entering the FM states, $\rho_{xx}(H)$ increases monotonically without additional inflection points (see Figure 4a). The high-field positive, non-saturating MR is likely dominated by orbital magnetoresistance from high-mobility carriers and is consistent with the calculated topological semi-metallic state in FM phases. To further quantify the transport anisotropy, the angular-dependent magnetoresistance (AMR), defined as $\text{AMR} = \frac{[\rho_{xx}(\theta, H) - \rho_{xx}(0^\circ, H)]}{\rho_{xx}(0^\circ, H)} \times 100\%$, is shown in Figure 3d. At 2

K and selected fields, AMR exhibits a clear twofold symmetry with maxima along the a -axis and minima for b -axis. As the field increases, the AMR peak at 90° ($H \parallel a$) initially grows rapidly up to 0.3 T, then decreases continuously toward zero at higher fields, indicating diminishing low-field anisotropy as FM state becomes nearly isotropic. Figure 3e summarizes the field-dependent AMR at 90° with increasing temperature. The pronounced non-monotonic field dependence at 2 K (rapid increase to a maximum followed by continuous decrease) is gradually suppressed at higher temperatures and vanishes above T_N , yielding negligible AMR at 10 K. Insets of Figure 3e compare the AMR(H, T) map at 90° with the magnetization anisotropy

between a - and b -axis, $M_a(H,T) - M_b(H,T)$, revealing similar non-monotonic trends below T_N and underscoring the intimate correlation between AMR and anisotropic magnetic order within low-field range. By contrast, field rotation in the bc -plane yields relatively small AMR (see Figure S25), consistent with the nearly isotropic $M(H)$ response within bc -plane.

Unlike the conventional Fermi-surface-anisotropy-induced AMR observed in nonmagnetic materials, whose peak amplitude typically increases monotonically with field [60–62], the non-monotonic field-dependence of AMR in $\text{Eu}_3\text{In}_2\text{As}_4$ resembles that reported in narrow-gap AFM semiconductors [63–65], where exchange-driven band reconstruction plays a dominant role. In such systems, magnetic anisotropy controls the spin configuration, while magnetization process along specific field directions reduces or even closes the bandgap.

Following this framework, we attribute the AMR to the combined effects of anisotropic magnetic scattering and exchange-driven band reconstruction. In $\text{Eu}_3\text{In}_2\text{As}_4$, first-principles calculations predict that the electronic structure evolves continuously from a gapped AFM state to a gapless FM state with topological band crossings. Because the exchange-induced band shift is proportional to $M(H)$, the bandgap evolution is expected to follow the magnetic polarization process. Consequently, the close correspondence between the AMR and the magnetization anisotropy provides transport evidence that the low-energy electronic structure is continuously reconstructed by the exchange coupling between localized Eu^{2+} moments and itinerant carriers.

2.4 | Quantum Oscillations in FM Phase $\text{Eu}_3\text{In}_2\text{As}_4$

Shubnikov-de Haas (SdH) oscillations were analyzed to further elucidate the Fermi surface topology of $\text{Eu}_3\text{In}_2\text{As}_4$. As shown in Figure 4a, pronounced quantum oscillations emerge in $\rho_{xx}(H)$ above ~ 8 T, a field sufficient to fully saturate the spin polarization and stabilize the FM band structure through exchange coupling. In this regime, the Landau-level spacing exceeds disorder broadening, enabling clear observation of SdH oscillations. Upon rotating the field from b - to a -axis (with angle θ), SdH oscillations persist in $\rho_{xx}(H)$. Meanwhile, once $\text{Eu}_3\text{In}_2\text{As}_4$ enters FM states, $\rho_{xx}(H)$ exhibits a quasi-linear, non-saturating increase, reminiscent of transport observations in Weyl systems [18, 66–71].

After subtracting non-oscillatory backgrounds via polynomial fitting, the oscillatory component $\Delta\rho_{xx}$ is plotted against $1/\mu_0 H$ (Figure 4b; curves are vertically offset by $1 \mu\Omega\cdot\text{cm}$). While oscillation amplitude remains largely unchanged, systematic phase evolutions occur with field direction. Owing to the limited oscillation window (~ 4 T), the oscillation frequency F was extracted from Landau fan diagrams (Figure S17a) rather than fast Fourier transform (FFT) analysis for its insufficient frequency resolution. A single dominant frequency ($F \sim 124$ T) is resolved for all directions, showing weak angular dependence (Figure 4c). Such behavior indicates a relatively simple and nearly isotropic three-dimensional (3D) Fermi surface in FM $\text{Eu}_3\text{In}_2\text{As}_4$, consistent with the simple-band-structure picture predicted by calculations. Through Onsager's relation, $F = (\hbar/2\pi e)S_F$, the frequency corresponds to the extremal cross-sectional area S_F of the Fermi

pocket perpendicular to the field, yielding $S_F = 1.18 \text{ nm}^{-2}$. Based on this S_F , we compute the 3D Fermi surface for different magnetization directions (Figure 4d), which remains close to an isotropic ellipsoid under magnetization rotation, consistent with the nearly angle-independent single-frequency SdH response. The experimental Fermi level is determined by matching the extracted S_F to the calculated Fermi surface cross-sectional area, yielding a value approximately 0.155 eV above the DFT Fermi energy, placing it above the Lifshitz transition. Besides, the large electron pocket aligns with the high electron density ($n_e = 2.1 \times 10^{19} \text{ cm}^{-3}$ and carrier mobility $\mu_e = 1015.1 \text{ cm}^2/\text{V}\cdot\text{s}$) derived from linear fitting of Hall signals (inset, Figure 4a), confirming that transport is dominated by a single high-mobility electron pocket.

Beyond the oscillation frequency, the SdH phase shift Φ is obtained from Landau-fan y -intercept, providing a complementary probe of the Berry-phase (ϕ_B) and band topology. Trivial parabolic bands yield $\phi_B = 0$, whereas linear band dispersions typically approach π -Berry phase. The extracted Φ values across field rotation are consistent with a π -Berry phase, supporting the presence of topologically nontrivial bands in the FM- $\text{Eu}_3\text{In}_2\text{As}_4$ under rotating magnetization (see Figure S17b).

In the fully polarized FM regime, $\text{Eu}_3\text{In}_2\text{As}_4$ behaves as a soft magnet whose spin moments continuously follow the field direction, rendering its band topology intrinsically angle-tunable. First-principles calculations predict a minimal Weyl platform: FMa (FMb) state supports an ideal Weyl semimetal with a single Weyl pair along Γ -X (Γ -Y). Upon rotating the magnetization within the ab -plane, canted-spin calculations show that the original Γ -Y Weyl crossing in FMb progressively gaps out while a gap simultaneously closes along Γ -X, culminating in a new Weyl pair in FMa limit (Figure S18). Throughout this rotation, the system remains Weyl semi-metallic. To visualize the Weyl cones, we calculated 3D band structures for a series of canting angles (Figure S19); a representative result at $\theta = 45^\circ$ is shown in Figure 4e, revealing a single Weyl-node pair. Figure 4f summarizes the calculated node trajectories, showing migration from high-symmetry lines to general k -points in the k_x - k_y plane, providing a direct handle to steer their momentum-space positions. As an independent verification, the Weyl-node search using WannierTools likewise identifies one Weyl pair residing in the $k_z = 0$ plane (Figure S19). By contrast, spin-rotation in the bc -plane largely preserves the ideal Weyl points along Γ -Y, until the FMc limit, where a nodal-ring semimetal forms in the k_x - k_y plane (Figure S18). Overall, $\text{Eu}_3\text{In}_2\text{As}_4$ enables rare vectorial tuning of a minimal Weyl phase via magnetization reorientation.

To further probe the topological electronic structure in the field-polarized state, we examined the anomalous Hall effect (AHE), which provides a sensitive transport probe of the Berry-curvature distribution and is quantified by anomalous Hall conductivity, σ_{AHE} . For an ideal type-I FM Weyl semimetal with the Fermi level located below the Lifshitz transition, the intrinsic σ_{AHE} exhibits Fermi-energy-independence and is only determined by the Weyl point separation Δk in momentum-space, $\sigma_{\text{AHE}} = (e^2/2\pi h)\Delta k$ [7]. In contrast, when the Fermi level is shifted away from the Weyl nodes or the Weyl cones become highly tilted (type-II), additional occupied-state and free-carrier contributions can introduce E_F

dependence [7, 8, 72, 73]. We extracted ρ_{AHE} by subtracting the linear ordinary Hall background from ρ_{yx} , followed by conversion to conductivity σ_{AHE} . Figure 4g shows $\sigma_{\text{AHE}}(H)$ with rotating fields. For all curves, σ_{AHE} remains negligible in AFM state, increases sharply across magnetic transitions toward FM order, and eventually saturates. The angular dependence of σ_{AHE} at 7 T (selected to avoid interference from SdH oscillations) is well-fitted by a cosine function (Figure 4h), primarily reflecting the geometric projection with field rotation. While band calculations indicate that magnetization rotation modifies Weyl point positions and separations, the absence of pronounced anomalies in angular-dependent σ_{AHE} suggests that the measured AHE is less sensitive to subtle Weyl cone evolution, likely due to the high electron density and the large Fermi pocket being away from the Weyl points.

To directly evaluate the Berry-curvature contribution at the experimentally relevant conditions, we calculated the intrinsic σ_{AHE} over a broad energy range in FM-Eu₃In₂As₄. As shown in Figure S21, the intrinsic σ_{AHE} is large near the Weyl nodes but remains finite at higher energies. At the experimentally determined Fermi level, the calculated $\sigma_{\text{ac,AHE}}$ for $H \parallel b$ is approximately $-10 \Omega^{-1}\text{cm}^{-1}$, which has the same sign and a comparable magnitude as the experimental value ($-28.62 \Omega^{-1}\text{cm}^{-1}$). This agreement indicates that Berry curvature associated with the calculated topological band structure contributes to the transport response, while the remaining quantitative discrepancy most likely originates from extrinsic contributions, such as skew-scattering and side-jump mechanisms. Scaling law analysis between the σ_{AHE} and σ_{xx} is also performed to distinguish intrinsic and extrinsic contributions (Note S8).

The evolution of SdH oscillations with field rotated in the bc -plane is shown in Figure S25. Moreover, temperature-dependent SdH oscillations (Figure S26) show that, while the oscillation amplitude is thermally damped, oscillations remain detectable even above T_{N} , consistent with FM correlations in PM regime that facilitate full spin polarization at high fields. More importantly, the extrema positions and oscillation frequency shift systematically with temperature. As shown in Figure S26c, the SdH frequency increases by $\sim 15\%$ from 2 to 20 K, which is anomalous for a rigid-band semimetal where the frequency is typically temperature-independent.

This frequency evolution clearly indicates a temperature-driven Fermi pocket reconstruction and can be naturally explained by the exchange-coupled nature of the Eu₃In₂As₄ band structure. The FM band structure is stabilized by exchange coupling to Eu²⁺ moments, thus, the thermal suppression of M_{sat} reduces the exchange-induced band splitting and modifies the extremal orbit area. A quantitative comparison between temperature evolution of the SdH frequency and the M_{sat} shows that the relative frequency shift, $F(T) - F(2 \text{ K})$, closely scales with the change in saturation magnetization, $M_{\text{sat}}(T) - M_{\text{sat}}(2 \text{ K})$ (Figure S26e). This strong correspondence suggests that the Eu²⁺ exchange field plays a dominant role in the Fermi-surface reconstruction. The qualitative derivation is detailed in Supporting Information Note S9.

Together with the magneto-infrared and AMR results presented above, these transport measurements consistently support the

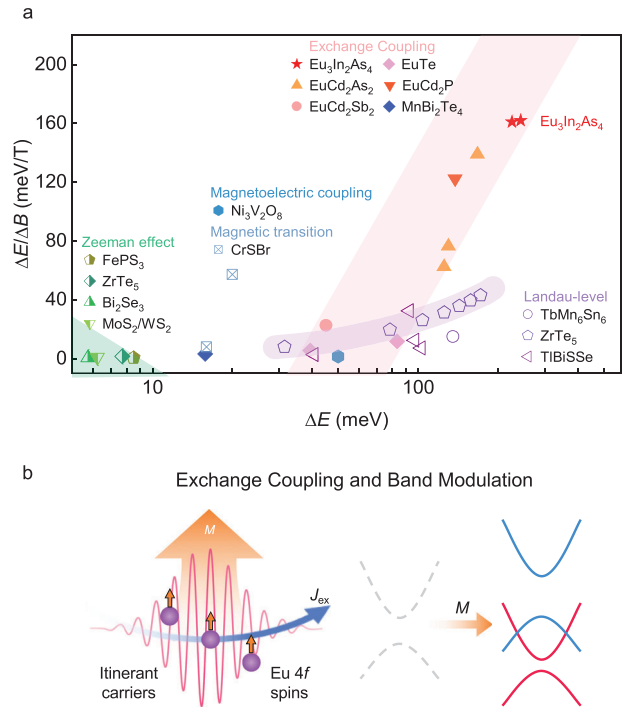


FIGURE 5 | The superiority of Eu²⁺-based exchange coupling in comparison with other mechanisms and its principle schematic. (a) Comparison of energy scales and effective average field slopes ($\Delta E/\Delta B$) across various modulation mechanisms: exchange coupling [23, 53–56], Zeeman effect [48, 50, 51, 103], magnetoelastic coupling [104], magnetic order transition [105], and Landau quantization [106–108]. (b) Illustration of the exchange coupling between localized spins and itinerant carriers. Purple balls refer to Eu²⁺ ions, and the electrons are illustrated as wave packet. The right panel demonstrates the M -induced exchange splitting and crossing, promoting the formation of Weyl points.

crucial role of exchange coupling in governing the electronic structure of Eu₃In₂As₄.

3 | Discussion

In contrast to the inter-site exchange interactions between localized magnetic moments that stabilize long-range magnetic order [74], we focus here on the exchange coupling between localized spins and itinerant carriers. Such spin-carrier coupling plays a central role in the emergence of topological electronic phases, including quantum anomalous Hall states [59], 1D Weyl modes [75], and 3D Van Hove singularities [55]. In Eu₃In₂As₄, we observe an unusually large exchange-driven band modulation, with characteristic energy shifts reaching ~ 300 meV. This energy scale is distinct from the bare exchange splitting commonly encountered in conventional ferromagnetic metals. Instead, it reflects a giant effective reconstruction of low-energy itinerant bands induced by exchange coupling in a weakly screened, low-carrier-density semimetal. Figure 5a compares the characteristic energy scales and effective average field slopes ($\Delta E/\Delta B$) across different band modulation mechanisms, compiled from magneto-optics, photoluminescence, or scanning tunneling microscopy studies. For saturating systems such as Eu₃In₂As₄, values correspond to saturation points; whereas for non-saturating cases,

maximum experimental energy shifts are used. Zeeman effect yields a weaker linear B -dependence of band splitting, observable only under high fields or in systems with a large g -factor. Landau level transitions display non-saturating, transition-index-dependent energy shifts. In contrast, exchange-driven band modulation exhibits both large energy scales and steep sub-saturation field dependence, particularly evident in $\text{Eu}_3\text{In}_2\text{As}_4$.

The origin of this strong exchange coupling in $\text{Eu}_3\text{In}_2\text{As}_4$ lies in the unique electronic configuration of Eu^{2+} . The half-filled $4f^7$ shell enforces, via Hund's rules, a maximal spin angular momentum ($S = 7/2$) and zero orbital angular momentum ($L = 0$). This yields a large and purely spin-derived local magnetic moment of $7 \mu_B$ as a giant internal exchange field, exceeding that typical of d -orbital systems. Crucially, the absence of L avoids higher-rank multipolar interactions associated with total angular momentum J , which are common in other $4f$ -magnets [58, 76]. This characteristic simplifies the exchange coupling mechanism to spin-only Heisenberg-form band splitting as $H_{\text{ex}} = -2J_{\text{ex}} \sum_i \mathbf{s}_i \cdot \mathbf{S}_i$ [55, 57–59], as the Equation (1) above. This form establishes a direct connection between the exchange-induced band splitting or shift and the macroscopic magnetization.

Furthermore, $L = 0$ renders Eu^{2+} insensitive to crystal electric fields, leading to isotropic and soft magnetism in $\text{Eu}_3\text{In}_2\text{As}_4$, enabling Eu moment saturation and reorientation under fields below ~ 1 T along different axes, essential for efficient vectorial control of the band topology: the external field rotates the Eu-spin polarization, while the exchange coupling transfers this rotation to the itinerant band structure. This sharply contrasts with $4f$ ions possessing nonzero L , where significant crystal-field splitting introduces pronounced magnetic anisotropy (e.g., Tb^{3+} in TbTi_3Bi_4 [77, 78], Dy^{3+} in quasi-Ising DyF_3 [79]), suppressing local moment along specific directions. It also differs from typical $3d$ magnets, where stronger anisotropy, orbital-hybridization-suppressed local moments, and defect-sensitive magnetic order make field control more difficult, such as MnBi_2Te_4 [80–82].

In addition, as a Zintl phase [83–86], $\text{Eu}_3\text{In}_2\text{As}_4$ features Eu^{2+} cations occupying the voids of the covalently bonded In-As anionic framework with well-defined valence, placing the magnetic moments in close proximity to the conductive channels and promoting appreciable exchange coupling with sizable J_{ex} . Meanwhile, the In-As framework yields a narrow bandgap with the conduction- and valence-band edges (In- s and As- p [45]) near E_F . The characteristics of low carrier density, narrow band gap, and sparse bands near E_F reduce metallic screening, allowing exchange-driven band shifts to play an outsized role in reshaping the topological properties. Together with the large isotropic Eu^{2+} moment, strong $4f$ localization, and the absence of crystal-field/hybridization-driven moment suppression, this simple band structure accounts for the giant exchange splitting in $\text{Eu}_3\text{In}_2\text{As}_4$, as illustrated in Figure 5b.

This capability is particularly impactful for realizing the ideal Weyl state, featuring a single pair of Weyl points, which remains experimentally rare. Most magnetic Weyl semimetals possess complex band structures with multiple Weyl cones or coexisting trivial Fermi pockets (e.g., $\text{Co}_3\text{Sn}_2\text{S}_2$ [10, 25]), obscuring intrinsic Weyl physics and often accompanied by reduced carrier mobility. Although ideal Weyl phases have been proposed in

several materials [4, 87–90], experimental realizations remain scarce such as FM- $\text{Mn}(\text{Bi}_{1-x}\text{Sb}_x)_2\text{Te}_4$ [72, 73] and $(\text{Cr,Bi})_2\text{Te}_3$ [90]. In $\text{Mn}(\text{Bi}_{1-x}\text{Sb}_x)_2\text{Te}_4$, moderate exchange coupling and strong magnetic anisotropy intrinsic to $3d$ magnetism require large fields to achieve FM polarization (~ 8 T along the c -axis and ~ 11 T in the ab -plane [80]) and antisite defects further suppress both magnetic and topological properties, even driving a degradation to trivial ferromagnetic insulators at high defect densities [42]. Other candidates, such as $\text{K}_2\text{Mn}_3(\text{AsO}_4)_3$ and EuCd_2As_2 , are challenged by recently reported semiconducting or insulating behavior [53, 91–93].

Beyond its nontrivial band topology, DFT calculations further reveal the altermagnetism in AFM- $\text{Eu}_3\text{In}_2\text{As}_4$, enriching the material's phase landscape. Distinct spin-split bands emerge along Γ -S and O-B paths, originating from the crystal symmetry associated with oppositely spin-polarized sublattices (see Figure S29).

4 | Conclusions

In summary, $\text{Eu}_3\text{In}_2\text{As}_4$ emerges as a uniquely efficient and versatile platform for exploring exchange-driven band engineering and magnetic topological phases, combining large band tunability, high carrier mobility, and exceptional magnetic controllability. The exchange-driven band modulation is directly visualized by large, magnetization-dependent optical-transition energy shifts, reaching ~ 300 meV within 3 T. Strong coupling to Eu^{2+} $4f$ moments yields a rich phase diagram, spanning an AFM a topological insulator ground state, magnetization-direction-tunable ideal FM Weyl semimetals, an ideal nodal-ring semimetal in FMC state, and a previously unexplored $2/3$ -FiMa phase. Although the specific topological assignments are based primarily on first-principles calculations, magnetization, AMR, SdH oscillations, AHE measurements and magneto-infrared spectroscopy collectively support exchange-driven band reconstruction across these magnetic phases. Direct spectroscopic observation of the Weyl cones and transport experiments closer to charge neutrality remain important goals for future work, calling for the Fermi-level tuning via optimized growth/chemical substitution, and on low-dimensional devices. Notably, $\text{Eu}_3\text{In}_2\text{As}_4$ nanowires have been synthesized via topotaxial mutual-exchange method [46], underscoring the device feasibility for gate-tunable platforms to access transport evidence of exotic boundary states.

5 | Methods

5.1 | Crystal Growth and Characterization

High-quality $\text{Eu}_3\text{In}_2\text{As}_4$ crystals were synthesized by the flux method [94]. Starting materials of high-purity Eu, In, As, and Sn (molar ratio 3: 15: 4.2: 15) were mixed into an alumina crucible fitted with an alumina filter within an Ar-filled glovebox (H_2O and $\text{O}_2 < 0.01$ ppm). The crucible was flame-sealed under vacuum in a quartz ampule. Heating was performed in a box furnace (KSL-1200X-J, Kejing) with the following temperature profile: ramp to 1000°C in 24 h, hold for 48 h to ensure homogenization and reaction, slowly cool to 800°C at 1°C/h , and dwell for 24 h. Crystals were subsequently separated from the excess

In-Sn flux via centrifugation. This method yielded $\text{Eu}_3\text{In}_2\text{As}_4$ crystals reaching several millimeters in length. Crystal structure and surface orientation were characterized by XRD using a Bruker D8 Discover diffractometer with $\text{Cu K}_{\alpha 1}$ radiation ($\lambda = 1.54056 \text{ \AA}$), achieved using a 2-bounce crystal monochromator to eliminate $\text{K}_{\alpha 2}$ component. Elemental composition was verified by EDX (Oxford X-Max). Room-temperature Raman spectrum was measured using a home-built system with 632.8 nm He-Ne laser excitation to detect Raman-active phonon modes.

5.2 | Magnetization and Electrical Transport Measurements

Magnetization measurements were performed using VSM mode in a magnetic property measurement system (MPMS3, Quantum Design, $\pm 7 \text{ T}$) with a standard quartz sample holder. For electrical transport measurements, needle-shaped samples were first polished to eliminate the residual In-Sn flux on the surface which are superconducting at low temperatures. Aluminum wires were bonded to the surface using silver paste (DuPont, 4929N) to form a standard six-wire Hall configuration. The magneto-transport measurements were performed using a Cryogenic superconducting magnet ($1.6 \sim 300 \text{ K}$, $\pm 12 \text{ T}$). A voltage-controlled current source (Stanford Research System, CS580) was used to provide an alternating current at 17.777 Hz, and the voltage signals were recorded by several lock-in amplifiers (Stanford Research System, SR865).

5.3 | Magneto-Infrared Measurement

Magneto-infrared spectroscopy was conducted at East China Normal University (ECNU, Shanghai) based on a home-built system consisting of a Fourier-transform-infrared (FTIR) spectrometer and a 12 T closed-cycle superconducting magnet [95]. The collimated infrared beam from a globar/tungsten light source was directed into the variable temperature insert (VTI) of the magnet and focused onto the (100) surface of the $\text{Eu}_3\text{In}_2\text{As}_4$ crystal using a parabolic mirror. The sample was maintained at approximately 6 K using a small amount of helium exchange gas. The reflected infrared beam was guided out of the VTI through a gold-coated brass light pipe and collected by a liquid-nitrogen-cooled Indium Antimonide (and Mercury Cadmium Telluride) detector in a detector chamber away from the magnet.

5.4 | Theoretical Calculations

The geometry and electronic structures of the studied system were calculated by using the projector augmented-wave (PAW) [96] formalism based on density functional theory (DFT), as implemented in the Vienna Ab initio Simulation Package (VASP) [97]. The Perdew–Burke–Ernzerhof generalized gradient approximation (GGA-PBE) [98] was employed to describe the exchange and correlation functional. The kinetic energy cutoff of the plane-wave basis was set to 550 eV, with a total energy convergence criterion of 10^{-6} eV . The Brillouin zone integration was performed by using an $8 \times 4 \times 12$ Γ -centered k point mesh. All atoms in the unit cell were allowed to relax until the Hellmann-Feynman force on each atom was less than $0.001 \text{ eV/\AA}$. To treat the correlation

effect of localized $4f$ electrons of Eu, the DFT+ U method [99] was employed with $U_{\text{eff}} = 7 \text{ eV}$. The phonon band structure was calculated using the density functional perturbation theory (DFPT) [100]. To investigate the topological properties, the maximally localized Wannier functions (MLWFs) were constructed by using the WANNIER90 package [101]. The edge states, 3D bands, the search for Weyl points and the energy-dependent anomalous Hall conductivity were calculated using the WANNIERTOOLS package [102].

Author Contributions

C.Z. conceived the idea and supervised the overall research. H.C. synthesized $\text{Eu}_3\text{In}_2\text{As}_4$ single crystals and performed structural and compositional characterizations. H.C. conducted magnetization measurements and analyzed the data with input from F.Z. X.D. and T.Z. carried out the first-principles calculations. G.W. performed magneto-infrared spectroscopy with assistance from Y.D., W.W., and X.Y. H.C. performed electrical transport measurements with the help of J.W., Z.X., Y.X., J.G., and P.L. G.W. and Y.D. performed Raman measurements on $\text{Eu}_3\text{In}_2\text{As}_4$. H.C. and C.Z. analyzed and interpreted the results with contributions from H.L., P.L., L.M., F.Z., T.Z. and X.Y. H.C. and C.Z. wrote the paper with assistance from all other co-authors.

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Conflicts of Interest

The authors declare no competing interests.

Data Availability Statement

All raw and derived data supporting the findings in the study are available from the corresponding author upon reasonable request.

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Supporting Information

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